Lecture 5

The particle in a box

Objectives. *Part I kept every system finite. A particle needs infinitely many dimensions. This lecture builds the answers of Part I for the simplest particle, which moves freely between two walls. Its states are represented by sequences of infinitely many amplitudes. Its Hamiltonian cannot act on every state. It is an operator with a* domain*, and we prove that it is self-adjoint there.*

5.1 Why a particle needs infinitely many dimensions

A particle moving along a line has a position and a momentum. In quantum mechanics, the observables Q for the position and P for the momentum satisfy the *canonical commutation relation*

$$QP\psi - PQ\psi = \mathrm{i}\hbar\,\psi \tag{5.1}$$

for every ψ in a suitable dense domain. Lecture 7 meets two such operators on the line, with their domain. Here we ask whether the spaces of Part I can hold them. In $\mathbb{C}^n$ the domain is the whole space, and (5.1) reads $QP - PQ = \mathrm{i}\hbar\mathbb{1}$.

Proposition 5.1 (No canonical pair of matrices)**.** *For every $n \geq 1$, there are no complex $n \times n$ matrices Q and P with $QP - PQ = \mathrm{i}\hbar\mathbb{1}$.*

Proof. The trace of a product does not depend on the order of the factors:

$$\mathrm{Tr}(QP) = \sum_{j,k} Q_{jk}P_{kj} = \sum_{k,j} P_{kj}Q_{jk} = \mathrm{Tr}(PQ).$$

So $\mathrm{Tr}(QP - PQ) = 0$. The trace of $\mathrm{i}\hbar\mathbb{1}$ is $\mathrm{i}\hbar n$, which is not 0. ∎

This is *Linear Algebra*, Prop. 14.24. A particle therefore needs a space of infinite dimension. The argument does not say which space. That choice is a postulate, made below for a particle confined to an interval.

The simplest such particle moves freely between two impenetrable walls, at $x = 0$ and $x = L$. It is the *particle in a box*.

5.2 Wavefunctions in a box

A state of the particle assigns an amplitude $\psi(x)$ to each position x in the box. The function ψ is the *wavefunction* of the state. Lecture 6 shows that $|\psi(x)|^2$ is the probability density for finding the particle near x. The wavefunctions form a vector space with the inner product

$$\langle \varphi, \psi \rangle = \int_0^L \overline{\varphi(x)}\,\psi(x)\,\mathrm{d}x, \tag{5.2}$$

which is antilinear in the first slot, as in Lecture 1. The space is $L^2(0, L)$. Its elements are the functions with $\int_0^L |\psi|^2 \, dx < \infty$, and two functions that differ only on a set of zero length count as the same element (*Linear Algebra*, Def. 13.19). With the norm $\|\psi\| = \langle \psi, \psi \rangle^{1/2}$, the space $L^2(0, L)$ is a Hilbert space, a complete inner product space.

So an element of $L^2(0, L)$ has no value at a single point. Changing ψ at $x = 0$ alone changes no integral, and it does not change the element. A condition such as $\psi(0) = 0$ therefore cannot be imposed on the states. It will return as a condition on the domain of the Hamiltonian.

Postulate 5.2 (States of a particle in a box)**.** The states of a particle in the box are the rays of $L^2(0, L)$.

Every state is represented by a sequence of amplitudes, the coefficients of a unit vector along the functions

$$u_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \qquad (n = 1, 2, 3, \dots). \tag{5.3}$$

The functions u_n are orthonormal, by the integral of a product of two sines (Exercise Sheet 5).

Theorem 5.3 (The sine basis · *cited*)**.** *The functions u_n form a complete orthonormal basis of $L^2(0, L)$. Every $\psi \in L^2(0, L)$ is the norm-convergent series*

$$\psi = \sum_{n=1}^{\infty} c_n u_n, \qquad c_n = \langle u_n, \psi \rangle \,, \tag{5.4}$$

and every sequence (c_n) with $\sum_n |c_n|^2 < \infty$ is the coefficient sequence of exactly one ψ. If $a_n = \langle u_n, \varphi \rangle$, then

$$\langle \varphi, \psi \rangle = \sum_{n=1}^{\infty} \overline{a_n} \, c_n \quad \text{and} \quad \|\psi\|^2 = \sum_{n=1}^{\infty} |c_n|^2 \,. \tag{5.5}$$

This is the Fourier sine series (Hall, Prop. 3.24). The general facts about complete bases are *Linear Algebra*, Thms 13.32 and 13.33, and the second identity in (5.5) is *Parseval's identity*. So a state of the box is represented by a sequence of infinitely many amplitudes $c_1, c_2, \dots$ with $\sum_n |c_n|^2 = 1$, up to a common phase, and every such sequence represents a state.

5.3 Stationary waves

What is the energy of the particle? For a free particle of mass m, Schrödinger's rule is that the energy acts on a wavefunction by $\psi \mapsto -(\hbar^2/2m)\, \psi''$. We take the rule as given. Lecture 7 explains it from the momentum on the line. A wavefunction of definite energy E, a *stationary wave*, solves $-(\hbar^2/2m)\, u'' = Eu$. Impenetrable walls are modelled by asking the stationary wave to vanish at both walls. Hall (§3.9) motivates this condition by a very deep well.

Lemma 5.4 (Stationary waves · *sketched*). *Let E be a real number. A nonzero, twice continuously differentiable u on $[0, L]$ with*

$$-\frac{\hbar^2}{2m}u'' = Eu \qquad \text{and} \qquad u(0) = u(L) = 0 \tag{5.6}$$

exists if and only if E is one of the energies

$$E_n = \frac{\hbar^2\pi^2 n^2}{2mL^2} \qquad (n = 1, 2, 3, \dots). \tag{5.7}$$

For $E = E_n$, the solutions are the multiples of u_n.

Proof. The equation is linear with constant coefficients, and its solutions form a two-dimensional space. If $E > 0$, put $k = \sqrt{2mE}/\hbar$. The solutions are $u = a\sin kx + b\cos kx$, with complex constants a and b. First, $u(0) = b$, so $b = 0$. Next, $u(L) = a\sin kL$, and a nonzero solution needs $\sin kL = 0$. With $k > 0$ this means $kL = n\pi$ for some $n \geq 1$, that is, $E = \hbar^2k^2/2m = E_n$, and then u is a multiple of u_n. For $E \leq 0$ the same two steps leave only $u = 0$, as Exercise Sheet 5 shows. ∎

The stationary waves are the sine functions u_n of the sine basis (Figure 5.1, left). A classical particle in the box may have any energy $E \geq 0$. The quantum particle has only the energies E_n, and even the least of them, $E_1 = \hbar^2\pi^2/2mL^2$, is positive.

5.4 Operators with domains

The energies grow without bound, $E_n = n^2E_1$, and this is where the new difficulty comes from. In the coefficients of (5.4), an energy that multiplies each stationary wave u_n by E_n multiplies each amplitude c_n by E_n. For some states the result is not square-summable, so it is the coefficient sequence of no vector of $L^2(0, L)$. § 5.7 gives such a state, with $c_n \propto 1/n^2$.

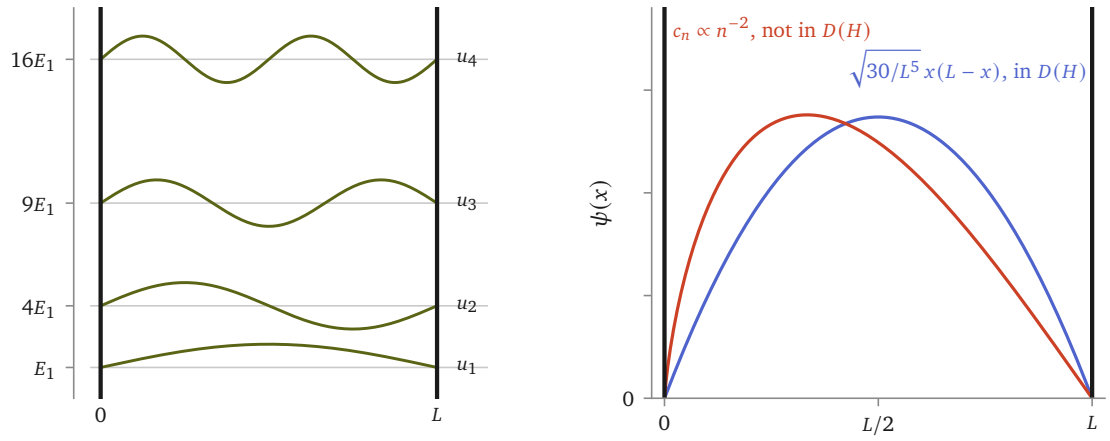


Figure 5.1: Left: the stationary waves $u_1, \dots, u_4$ between the walls, each drawn about the height of its energy $E_n = n^2E_1$. The vertical size of each wave is not to scale. Right: the state ψ of Example 5.13, which lies in the domain of H, and the state with $c_n \propto n^{-2}$ of § 5.7, which does not.

In Part I every observable was a matrix, and a matrix acts on every vector. In infinite dimension, an operator comes with the set of vectors it acts on.

Definition 5.5 (Operator; densely defined)**.** An *operator* A on a Hilbert space $\mathcal{H}$ is a linear map from a subspace $D(A) \subseteq \mathcal{H}$, its *domain*, into $\mathcal{H}$. It is *densely defined* if every vector of $\mathcal{H}$ is the limit of a sequence in $D(A)$. Two operators are equal when they have the same domain and the same action on it.

Definition 5.6 (Adjoint)**.** Let A be a densely defined operator on $\mathcal{H}$. Its *adjoint* $A^\dagger$ has the domain

$$D(A^\dagger) = \big\{\varphi \in \mathcal{H} : \text{ there is } \chi \in \mathcal{H} \text{ with } \langle \varphi, A\psi\rangle = \langle \chi, \psi\rangle \text{ for every } \psi \in D(A)\big\}, \tag{5.8}$$

and on it $A^\dagger\varphi = \chi$.

The vector χ is unique, because $D(A)$ is dense (Exercise Sheet 5). On $\mathbb{C}^n$ with $D(A) = \mathbb{C}^n$, the adjoint is the conjugate transpose, as in Part I. In infinite dimension the vector χ need not exist for every φ, and $D(A^\dagger)$ records the φ for which it does.

Definition 5.7 (Symmetric; self-adjoint)**.** A densely defined operator A is *symmetric* if $\langle \varphi, A\psi\rangle = \langle A\varphi, \psi\rangle$ for all $\varphi, \psi \in D(A)$. It is *self-adjoint* if $A^\dagger = A$, that is, if $D(A^\dagger) = D(A)$ and $A^\dagger\varphi = A\varphi$ there.

If A is symmetric and $\varphi \in D(A)$, then $\chi = A\varphi$ satisfies the condition in (5.8). So $D(A) \subseteq D(A^\dagger)$, with $A^\dagger\varphi = A\varphi$ there. Self-adjointness asks in addition that the adjoint have no larger domain. On $\mathbb{C}^n$ the two notions coincide, and both are the self-adjoint matrices of Part I.

Postulate 5.8 (Observables with an eigenbasis)**.** The observables of a particle in the box are self-adjoint operators on $L^2(0, L)$. Suppose an observable A has an orthonormal basis of eigenvectors $e_1, e_2, \ldots$, with $Ae_k = a_k e_k$. Then its readings are the numbers a_k, and the measurement postulate of Lecture 2 holds, with

$$E_a\psi = \sum_{k:\, a_k = a} \langle e_k, \psi\rangle\, e_k, \qquad \text{Prob}(a) = \|E_a\psi\|^2 = \sum_{k:\, a_k = a} |\langle e_k, \psi\rangle|^2 . \tag{5.9}$$

By Parseval's identity, the probabilities add to 1. This is where completeness matters. If the basis missed a direction, part of the probability would be lost.

5.5 The box Hamiltonian

Postulate 5.9 (The energy in a box)**.** The energy of a particle of mass m in the box is an observable H. On every twice continuously differentiable ψ on $[0, L]$ with $\psi(0) = \psi(L) = 0$, it acts by Schrödinger's rule

$$H\psi = -\frac{\hbar^2}{2m}\,\psi''. \tag{5.10}$$

The postulate fixes H on a small set of functions. Does it fix H? We first build one self-adjoint operator that obeys it, from the coefficients.

Definition 5.10 (The box Hamiltonian)**.** With $c_n = \langle u_n, \psi \rangle$, the *box Hamiltonian* H has the domain and the action

$$D(H) = \left\{ \psi \in L^2(0, L) : \sum_{n=1}^{\infty} n^4 |c_n|^2 < \infty \right\}, \qquad H\psi = \sum_{n=1}^{\infty} E_n c_n u_n. \tag{5.11}$$

Since $E_n = n^2 E_1$, the condition on the domain says $\sum_n E_n^2 |c_n|^2 < \infty$. By the sine basis theorem, this is exactly the condition for the series $\sum_n E_n c_n u_n$ to converge. In the coefficients, H is the diagonal operator $(c_n) \mapsto (E_n c_n)$.

Theorem 5.11 (The box Hamiltonian)**.** *The box Hamiltonian H is densely defined and self-adjoint. Its eigenvalues are exactly the energies E_n, and the eigenvectors for E_n are the nonzero multiples of u_n. It obeys the energy postulate. Every twice continuously differentiable ψ on $[0, L]$ with $\psi(0) = \psi(L) = 0$ lies in $D(H)$, and $H\psi = -(\hbar^2/2m)\,\psi''$.*

Proof.

1. *Dense.* A finite combination of the u_n lies in $D(H)$, since its sum $\sum_n n^4 |c_n|^2$ has finitely many nonzero terms. By the sine basis theorem, the partial sums $\sum_{n \le N} c_n u_n$ of any ψ converge to ψ. So $D(H)$ is dense.
2. *Symmetric.* Let $\varphi, \psi \in D(H)$, with coefficients a_n and c_n. Since each E_n is real, (5.5) gives
$$\langle \varphi, H\psi \rangle = \sum_n \overline{a_n}\, E_n c_n = \sum_n \overline{E_n a_n}\, c_n = \langle H\varphi, \psi \rangle .$$
3. *No larger adjoint.* Let $\varphi \in D(H^\dagger)$, with $H^\dagger \varphi = \chi$, and put $a_n = \langle u_n, \varphi \rangle$. Take $\psi = u_n$ in (5.8). The left side is $\langle \varphi, Hu_n \rangle = E_n \langle \varphi, u_n \rangle = E_n \overline{a_n}$, and the right side is $\langle \chi, u_n \rangle = \overline{\langle u_n, \chi \rangle}$. So $\langle u_n, \chi \rangle = E_n a_n$ for every n. By Parseval's identity, $\sum_n E_n^2 |a_n|^2 = \|\chi\|^2 < \infty$. Hence $\varphi \in D(H)$ and $\chi = \sum_n E_n a_n u_n = H\varphi$. So $D(H^\dagger) \subseteq D(H)$. By the symmetric step and the remark after Definition 5.7, the opposite inclusion holds, with $H^\dagger \varphi = H\varphi$. So $H^\dagger = H$.
4. *Eigenvalues.* By definition, $Hu_n = E_n u_n$. Conversely, let $H\psi = \lambda\psi$ with $\psi \neq 0$. Comparing coefficients gives $E_n c_n = \lambda c_n$ for every n. Some c_n is nonzero, so $\lambda = E_n$. The energies are distinct, so $c_j = 0$ for every $j \neq n$, and $\psi = c_n u_n$.
5. *The rule.* Let ψ be twice continuously differentiable on $[0, L]$, with $\psi(0) = \psi(L) = 0$. Since the u_n are real, integrating by parts twice gives
$$\langle u_n, -\psi'' \rangle = \left\langle -u_n'', \psi \right\rangle + \Big[u_n' \psi - u_n \psi' \Big]_0^L .$$
The bracket vanishes, because u_n and ψ both vanish at 0 and at L. Since $-u_n'' = (n\pi/L)^2 u_n$, the coefficients of $-(\hbar^2/2m)\,\psi''$ are $E_n c_n$. The function ψ'' is continuous on $[0, L]$, so it lies in $L^2(0, L)$, and Parseval's identity gives $\sum_n E_n^2 |c_n|^2 = (\hbar^2/2m)^2 \|\psi''\|^2 < \infty$. Hence $\psi \in D(H)$, and $H\psi = \sum_n E_n c_n u_n = -(\hbar^2/2m)\,\psi''$. ∎

So H obeys the energy postulate. Is it the only observable that does? Yes. A cited theorem says so, and it also describes the domain by derivatives. A function ψ on $[0, L]$ is *absolutely continuous* if $\psi(x) = \psi(0) + \int_0^x g(t)\,dt$ for an integrable function g. Then $\psi' = g$ except on a set of zero length, and ψ has a value at every point.

Theorem 5.12 (The Dirichlet domain · *cited*)**.** *The box Hamiltonian is the only self-adjoint*

operator that obeys the energy postulate. Its domain is

$$D(H) = \big\{\psi \in L^2(0,L) : \psi \text{ and } \psi' \text{ are absolutely continuous on } [0,L], \\ \psi'' \in L^2(0,L),\ \psi(0) = \psi(L) = 0\big\}, \tag{5.12}$$

and $H\psi = -(\hbar^2/2m)\,\psi''$ *for every* ψ *in it.*

Teta, Ex. 4.22, proves that the box Hamiltonian is the only one, with $\hbar = 1$, $m = \frac{1}{2}$ and $L = \pi$. Teschl, (2.57), gives its domain in derivatives. The condition $\psi(0) = \psi(L) = 0$ is the *Dirichlet condition*. It makes sense in (5.12) because an absolutely continuous function has a value at every point. The walls are a condition on the domain of H, not on the states. The energy postulate therefore determines the Hamiltonian.

5.6 Energy readings in a parabola

Example 5.13 (A parabola in the box). Figure 5.1, right, shows the state of this example.

1. *Prepare.* The state is
$$\psi(x) = \sqrt{\frac{30}{L^5}}\,x(L-x).$$
It vanishes at both walls. Since $\int_0^L x^2(L-x)^2\,\mathrm{d}x = L^5/30$, it is a unit vector.
2. *Find the coefficients.* The function ψ is twice continuously differentiable and vanishes at the walls. So the step on the rule in the proof of Theorem 5.11 gives $(n\pi/L)^2 c_n = \langle u_n, -\psi''\rangle$. Here $-\psi'' = 2\sqrt{30/L^5}$ is a constant, and
$$\langle u_n, 1\rangle = \sqrt{\frac{2}{L}}\int_0^L \sin\frac{n\pi x}{L}\,\mathrm{d}x = \sqrt{\frac{2}{L}}\,\frac{L\,(1-(-1)^n)}{n\pi}.$$
Hence
$$c_n = \frac{8\sqrt{15}}{\pi^3 n^3} \quad \text{for odd } n, \qquad c_n = 0 \quad \text{for even } n.$$
3. *Read the energies.* By the observables postulate, a measurement of the energy gives E_n with probability c_n^2, and it leaves the state u_n. So $\mathrm{Prob}(E_1) = 960/\pi^6 \approx 0.9986$ and $\mathrm{Prob}(E_3) = 960/(729\pi^6) \approx 0.0014$. The even levels never occur, because the state is symmetric about the midpoint $L/2$ and the u_n with even n are antisymmetric about it. By Parseval's identity the probabilities add to 1.
4. *Apply H.* By Theorem 5.11, $\psi \in D(H)$, and
$$H\psi = -\frac{\hbar^2}{2m}\,\psi'' = \frac{\hbar^2}{m}\sqrt{\frac{30}{L^5}},$$
a constant function. It does not vanish at the walls. The walls are a condition on ψ, not on $H\psi$.
5. *Find the mean energy.* By the inner product formula (5.5) and the previous step, the mean of the energy readings is
$$\sum_n E_n\,\mathrm{Prob}(E_n) = \langle \psi, H\psi\rangle = \frac{\hbar^2}{m}\sqrt{\frac{30}{L^5}}\int_0^L \psi(x)\,\mathrm{d}x = \frac{5\hbar^2}{mL^2} = \frac{10}{\pi^2}\,E_1 \approx 1.013\,E_1.$$
The mean lies just above the least energy E_1, as the probabilities in the third step predict.

5.7 A state outside the domain

Is every state in the domain of H? No. Take the coefficients

$$c_n = \frac{\sqrt{90}}{\pi^2 n^2} \qquad (n = 1, 2, 3, \dots).$$

By Euler's sum $\sum_n 1/n^4 = \pi^4/90$, the coefficients define a unit vector $\psi = \sum_n c_n u_n$. But $n^4 c_n^2 = 90/\pi^4$ for every n, so $\sum_n n^4 c_n^2$ diverges, and $\psi \notin D(H)$. The vector $H\psi$ is not defined.

The energy readings of ψ are nevertheless defined, by the observables postulate. By Euler's sum $\sum_n 1/n^2 = \pi^2/6$, the mean energy is

$$\sum_n E_n \operatorname{Prob}(E_n) = \frac{90}{\pi^4} E_1 \sum_n \frac{1}{n^2} = \frac{15}{\pi^2} E_1 \approx 1.52\, E_1,$$

which is finite. The domain is therefore not a set of allowed states. Every unit vector of $L^2(0, L)$ represents a state, its energy can be measured, and Lecture 6 gives its time evolution.

What does the domain mean, then? For $\psi \in D(H)$, Parseval's identity gives

$$\|H\psi\|^2 = \sum_n E_n^2 \, |c_n|^2 = \sum_n E_n^2 \operatorname{Prob}(E_n),$$

the mean square of the energy readings. So a state lies in $D(H)$ exactly when its energy readings have a finite mean square. For the state above the mean is finite and the mean square is not.

Why insist on self-adjoint, and not only symmetric? Physics texts often call an operator *Hermitian* when it is symmetric. Integration by parts, as in the step on the rule, proves that Schrödinger's rule is symmetric on the twice continuously differentiable functions that vanish at the walls. It does not prove self-adjointness, which also requires that the adjoint have no larger domain. In infinite dimension the two notions differ. Schrödinger's rule on those functions alone is an example. It is symmetric and not self-adjoint (Exercise Sheet 5). *Quantum Mechanics II*, Lecture 7, shows another operator in this box, the momentum with walls.

The particle in a box, as a quantum system

States: the rays of $L^2(0, L)$, represented by the sequences of coefficients $c_n = \langle u_n, \psi \rangle$ with $\sum_n |c_n|^2 = 1$, up to a common phase (Postulate 5.2 and Theorem 5.3). *Observables:* self-adjoint operators, each with its domain (Postulate 5.8 and Definition 5.7). *Energy:* H multiplies c_n by $E_n = \hbar^2 \pi^2 n^2 / 2mL^2$ on the states with $\sum_n n^4 |c_n|^2 < \infty$. It is self-adjoint and obeys Schrödinger's rule (Postulate 5.9 and Theorems 5.11 and 5.12). *Measurement:* the energy reads E_n with probability $|c_n|^2$, and the state afterwards is u_n (Postulate 5.8).

Summary.

- *The box Hamiltonian* is self-adjoint on its domain, its eigenvalues are the E_n, and it obeys Schrödinger's rule (Theorem 5.11). It is the only self-adjoint operator that does (Theorem 5.12).
- *A state outside the domain*: the domain holds the states whose energy readings have a finite mean square.

Lecture 6 lets the box Hamiltonian evolve the states, and asks where the particle is.

References

Teta, *A Mathematical Primer on Quantum Mechanics*, Ex. 4.20 and Prop. 4.17 (p. 100), Ex. 4.22 (pp. 101–102). Teschl, *Mathematical Methods in Quantum Mechanics*, (2.50)–(2.57) (pp. 69–70). Hall, *Quantum Theory for Mathematicians*, §3.9 (pp. 80–82) and §9.2 (pp. 170–172). Griffiths and Schroeter, *Introduction to Quantum Mechanics*, §2.2 (pp. 31–38). *Linear Algebra for Physics*, Def. 13.19, Thms 13.32 and 13.33, Prop. 14.24.